



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- Performed data collection via RESTful API calls to the public SpaceX API and web scraping of the SpaceX Wikipedia page
- Conducted Exploratory Data Analysis using SQL queries, Matplotlib/Seaborn visualizations, Folium geospatial mapping, and interactive Plotly Dash dashboards
- Executed feature standardization and performed hyperparameter tuning via GridSearchCV with k-fold cross-validation to optimize multiple classification models

- **Summary of all results**

- KSC LC-39A accounted for the highest successful launches with 41.7% of total successful landings across all sites
- Landing success rate has a positive upward trend from 2013 onward, surpassing an 80% success rate by 2020, showing improvements in booster recovery technology
- Among evaluated classifiers, the Decision Tree model achieved the highest predictive accuracy at 94.44%, outperforming SVM, KNN, and Logistic Regression, which scored less test set
- Feature importance analysis indicates that launch site and booster version are stronger predictive features for landing success than payload mass in isolation, suggesting categorical/operational factors outweigh continuous payload variables in model decision boundaries.

Introduction

- **Project background and context**

- SpaceX has transformed the launch industry by successfully recovering and reusing the Falcon 9 first-stage booster, driving launch costs down from \$165M+ to \$62M per launch. Being able to accurately predict first-stage landing success allows for reliable launch cost estimation, offering competing launch providers valuable competitive intelligence for strategic bidding.

- **Problem**

- This project seeks to answer three central questions: what operational and technical factors drive Falcon 9 first-stage landing success, which machine learning model most accurately predicts landing outcomes, and how variables such as payload mass, launch site, and orbit type influence the probability of a successful landing.

Section 1

Methodology

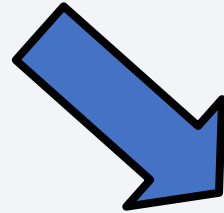
Methodology

Data collection methodology:

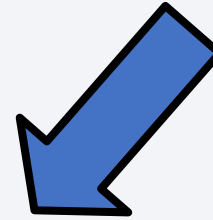
- Data was collected from two sources the SpaceX public REST API and web-scraped historical launch records from the SpaceX Wikipedia page to build a more complete and comprehensive dataset for analysis
- **Perform data wrangling**
 - Data cleaning addressed missing values (via imputation or removal), standardized inconsistent formatting across fields, and resolved data inconsistencies to ensure integrity
 - Feature extraction and engineering were performed to enrich the dataset, deriving new predictive variables from the raw data to enhance the depth of subsequent EDA and modeling
- **Perform exploratory data analysis (EDA) using visualization and SQL**
 - Visualized launch success rates, payload mass distributions, and launch site performance using Matplotlib and Seaborn, enabling identification of trends and patterns in the data
 - Executed targeted SQL queries to extract specific insights and answer key analytical questions about launch outcomes and dataset attributes
- **Perform interactive visual analytics using Folium and Plotly Dash**
 - Built interactive geospatial maps using Folium to visualize launch sites and their corresponding landing outcomes geographically
 - Developed an interactive Plotly Dash application featuring dropdown menus and range sliders
- **Perform predictive analysis using classification models**
 - Built and benchmarked Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Decision Tree to predict landing outcomes
 - Applied GridSearchCV with cross-validation to systematically identify the optimal hyperparameters for each model
 - Evaluated all classifiers on accuracy score and identified the top-performing model

Data Collection

SpaceX API Request



Web Scraping Wikipedia

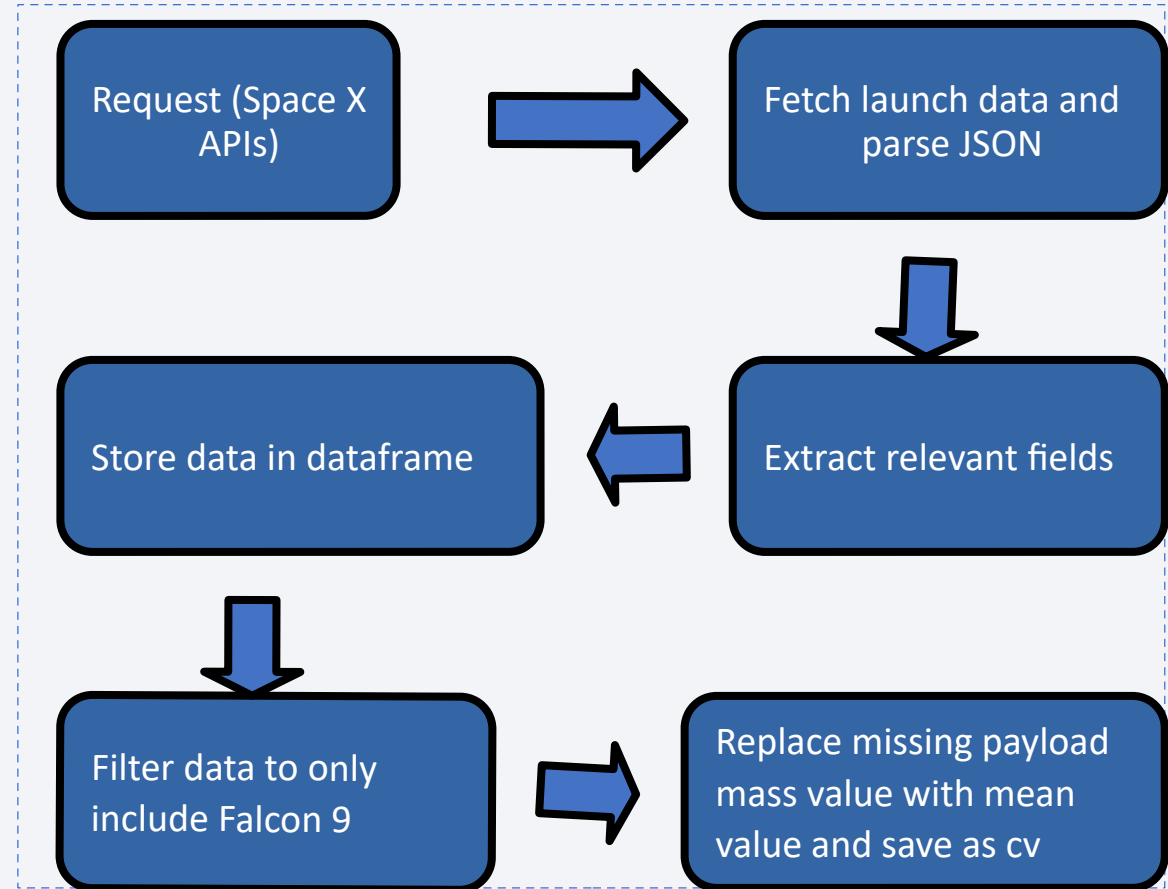


Data Integration

Data Collection – SpaceX API

Present your data collection with SpaceX REST calls using key phrases and flowcharts

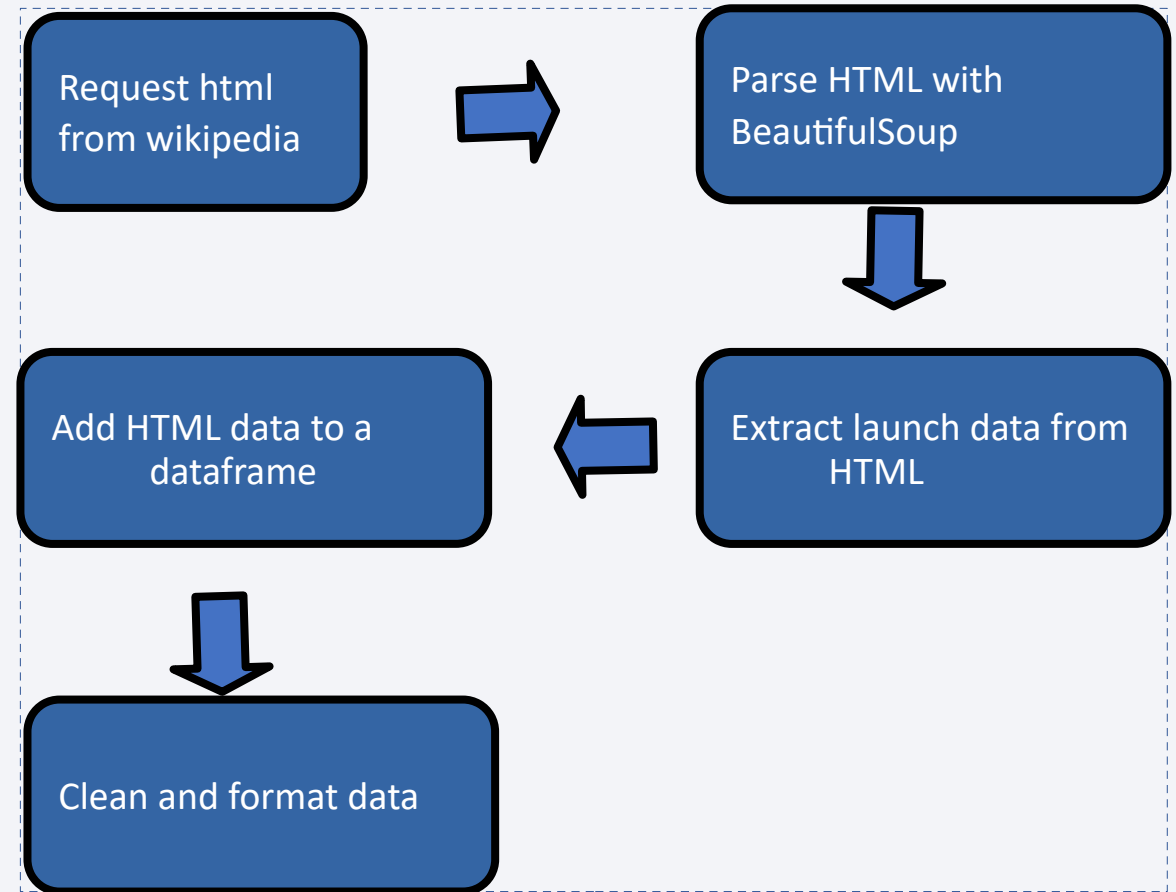
<https://github.com/RogerIO-git/IBM-Coursera-Capstone/blob/main/Module%201/jupyter-labs-spacex-data-collection-api-v2.ipynb>



Data Collection - Scraping

Falcon 9 launch records were web scraped from Wikipedia using the requests library for HTTP calls and BeautifulSoup for HTML parsing.

<https://github.com/RogerIO-git/IBM-Coursera-Capstone/blob/main/Module%201/jupyter-labs-webscraping.ipynb>



Data Wrangling

Step 1 - Data Cleaning

- Find missing values and either fill them in or remove them.
- Drop rows and columns that have too little data to use.

Step 2 - Data Transformation

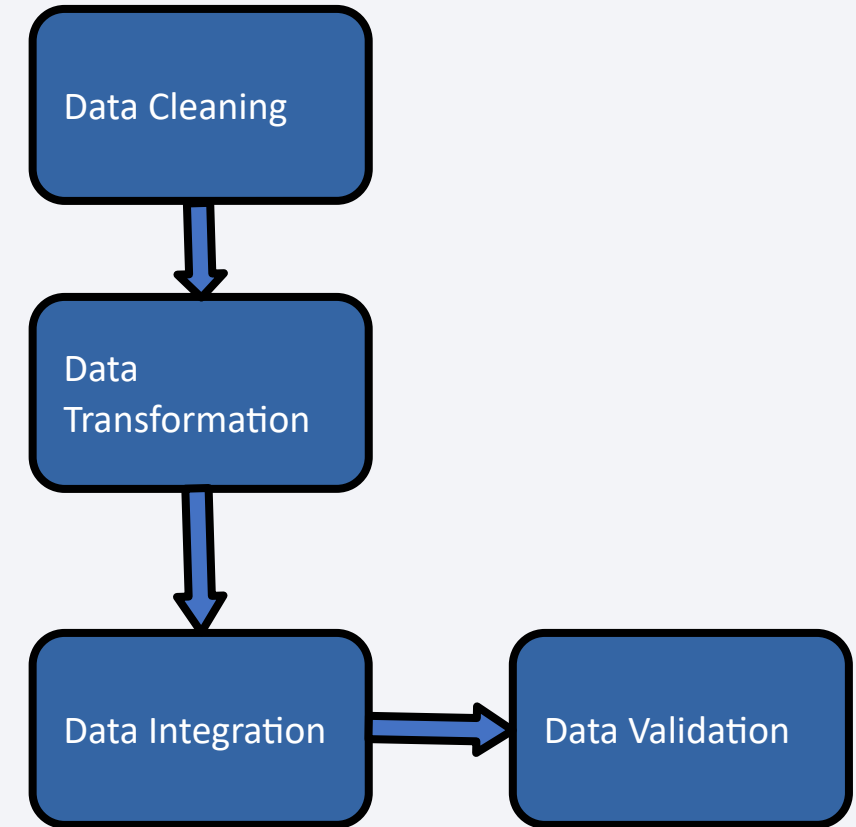
- Convert columns to the right data types.
- Normalize numerical features.
- Standardize text by converting to lowercase and stripping whitespace.

Step 3 - Data Integration

- Combine the API and scraped datasets into one cleaned table.

Step 4 - Data Validation

- Find and remove any duplicate records.



EDA with Data Visualization

Payload mass (kg) vs. Flight Number, Launch Site vs. Flight Number, Payload mass (kg) vs. Launch Site, Orbit vs. Flight Number and Orbit vs. Payload mass (kg) figures were shown using scatterplots

Success Rate vs. Orbit was shown using a **bar chart**

Average Success Rate vs. Year was shown using **line chart**

Bar Charts: To compare how launch outcomes stack up across categories like launch sites and rocket types. This makes it easy to see where success rates are higher or lower across different groups.

Line Charts: To track how things like success rate change year over year. This reveals trends over time and shows whether performance is improving or declining.

Scatter Plots: To explore how two numerical variables relate to each other, like payload mass vs. landing success. This uncovers correlations and shows how one variable moves in response to another.

GitHub: <https://github.com/RogerIO-git/IBM-Coursera-Capstone/blob/main/Module%202/eda-with-visualation.ipynb>

EDA with SQL

Aggregate Queries

- Counted the total number of launches.
- Sum successful and failed launches separately.
- Calculated success rates broken down by launch site and rocket type.

Join Queries

- Joined tables to connect launch records with rocket specs data.

Filtering Queries

- Filtered data for outcome types like success or failure.

Sorting Queries

- Sorted data to find trends and outliers.
- Ordered launches by date and success rate.

Subqueries

Build an Interactive Map with Folium

Markers

- At each launch site to identify them on the map.
 - Marker represents a real location where SpaceX launches take place.

Circles

- Drawn around launch sites to show proximity radius:
 - Circles help to show the areas around each site

Lines

- Connects launch sites to nearby key locations.
 - Lines add context and direct towards important location

GitHub: https://github.com/RogerIO-git/IBM-Coursera-Capstone/blob/main/Module%203/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Graphs:

Pie chart - Show distribution of successful landings across all launch sites.

Scatter plot - All sites or individual site and payload mass are shown.

Interactions:

Pie chart - This is used to visualize launch site success rate distribution among all successful launches.

Scatter plot - This help us see how payload weight connects to landing success. Payload mass on a slider between 0 and 10000 kg helps to dynamically change and see how change with change in payload mass.

GitHub: <https://github.com/RogerIO-git/IBM-Coursera-Capstone/blob/main/Module%203/space-dash-app.py>

Predictive Analysis (Classification)

Data Preprocessing:

- Standardized all features so no single variable carries too much weight as well as split the data into training and test sets.

Model Selection:

- SVM, Decision Trees, Logistic Regression and K-Nearest Neighbors were used.

Hyperparameter Tuning:

- Used GridSearchCV to find the best tuning.

Model Evaluation:

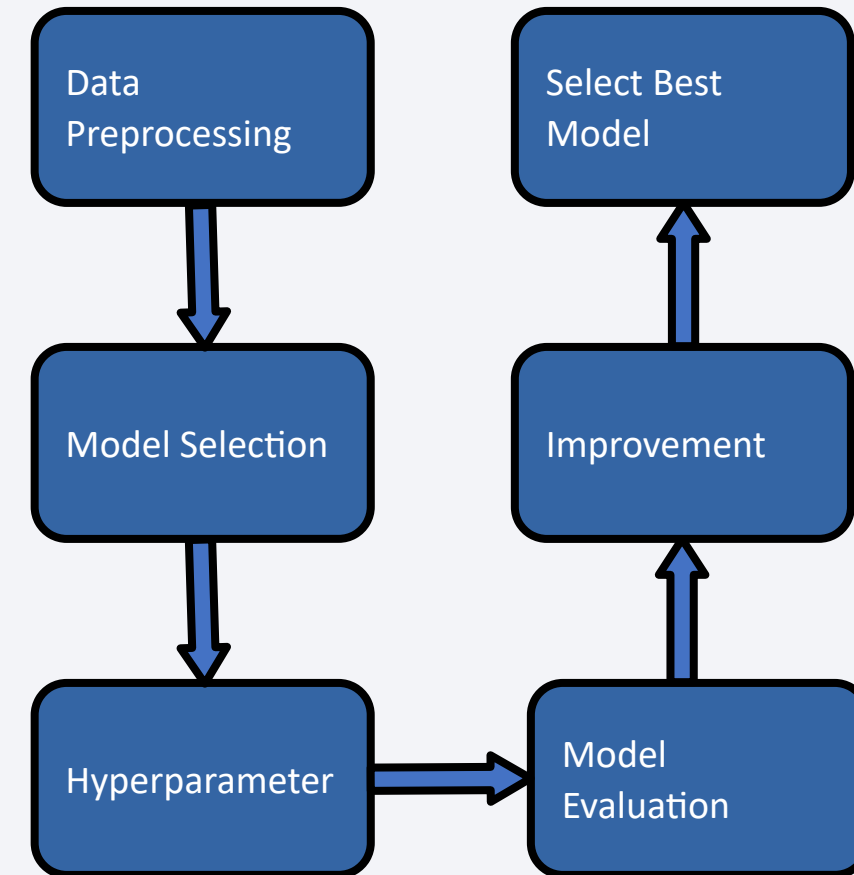
- Ran cross-validation to make sure results would hold up on new data. Measured accuracy, recall and precision of each model.

Improvement:

- Adjusted models based on what the validation results revealed.

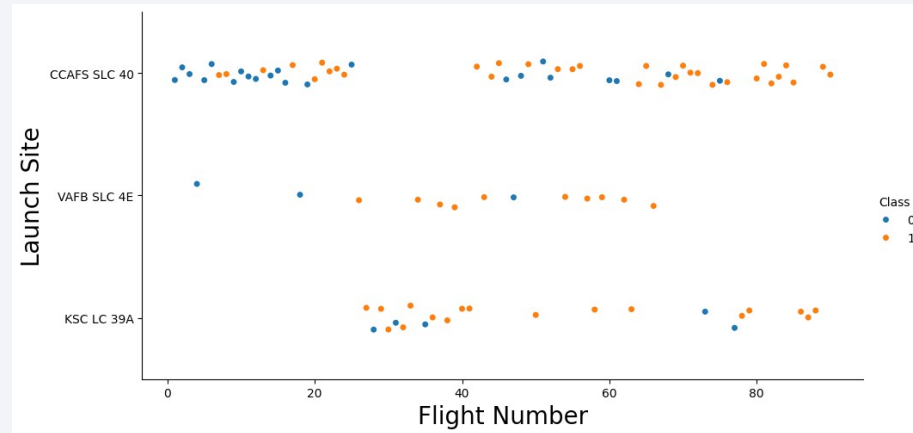
Select Best Model:

- Chose the model with the highest test set accuracy as the winner.

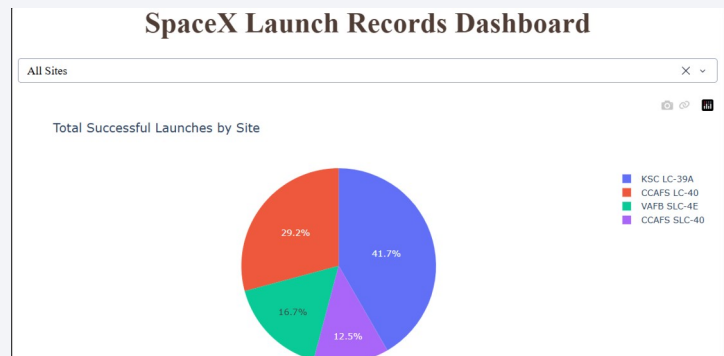


Results

- Exploratory data analysis results



- Interactive analytics demo in screenshots



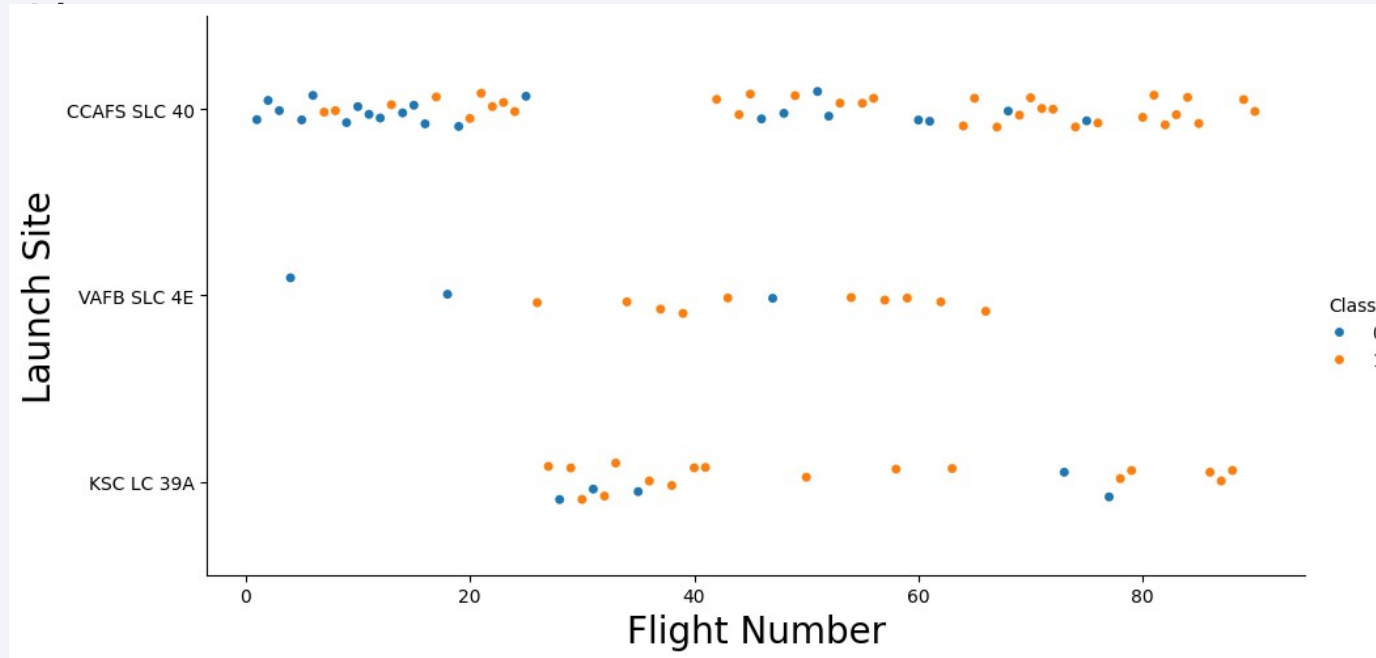
The background of the slide is a complex, abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks and lines in shades of red and cyan. These lines vary in thickness and opacity, creating a sense of depth and movement. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is a high-tech, digital aesthetic.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

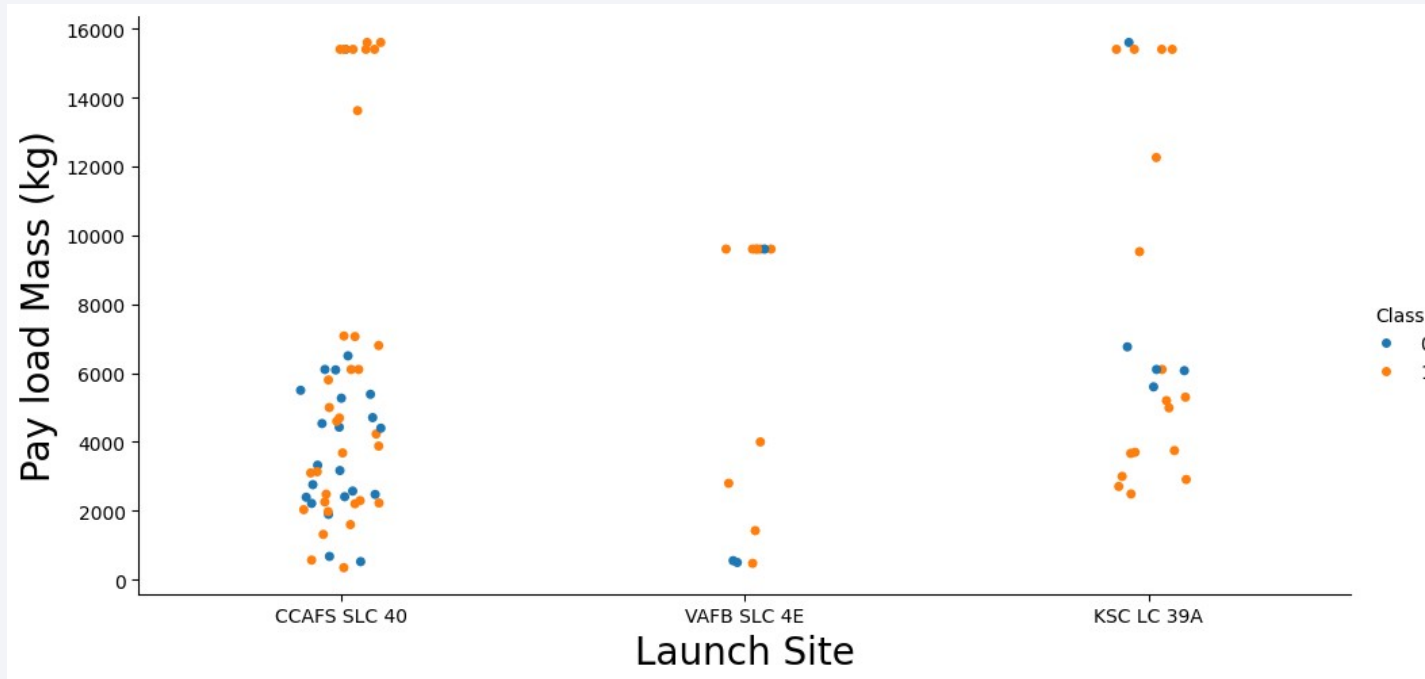
Flight Number vs. Launch



Yellow represents success while blue represents failure. The graph shows an increase in success rate over time. Failures didn't stop but not as much is observed consistently pass 20 flights. Launch site CCAFS SLC 40 seems to be the main site indicated by the higher volume of flights from there.

Payload vs. Launch Site

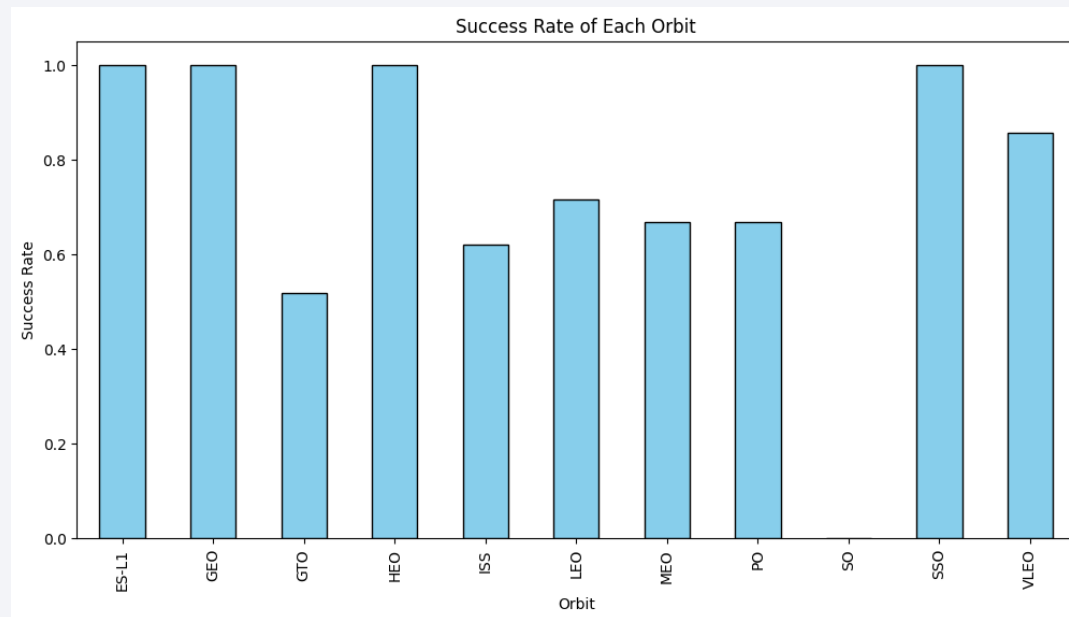
Payload vs. Launch Site



Yellow represents success while blue represents failure. The graph shows that launch site VAFB SLC 4E isn't used to for payload over 1000 kg.

Success Rate vs. Orbit Type

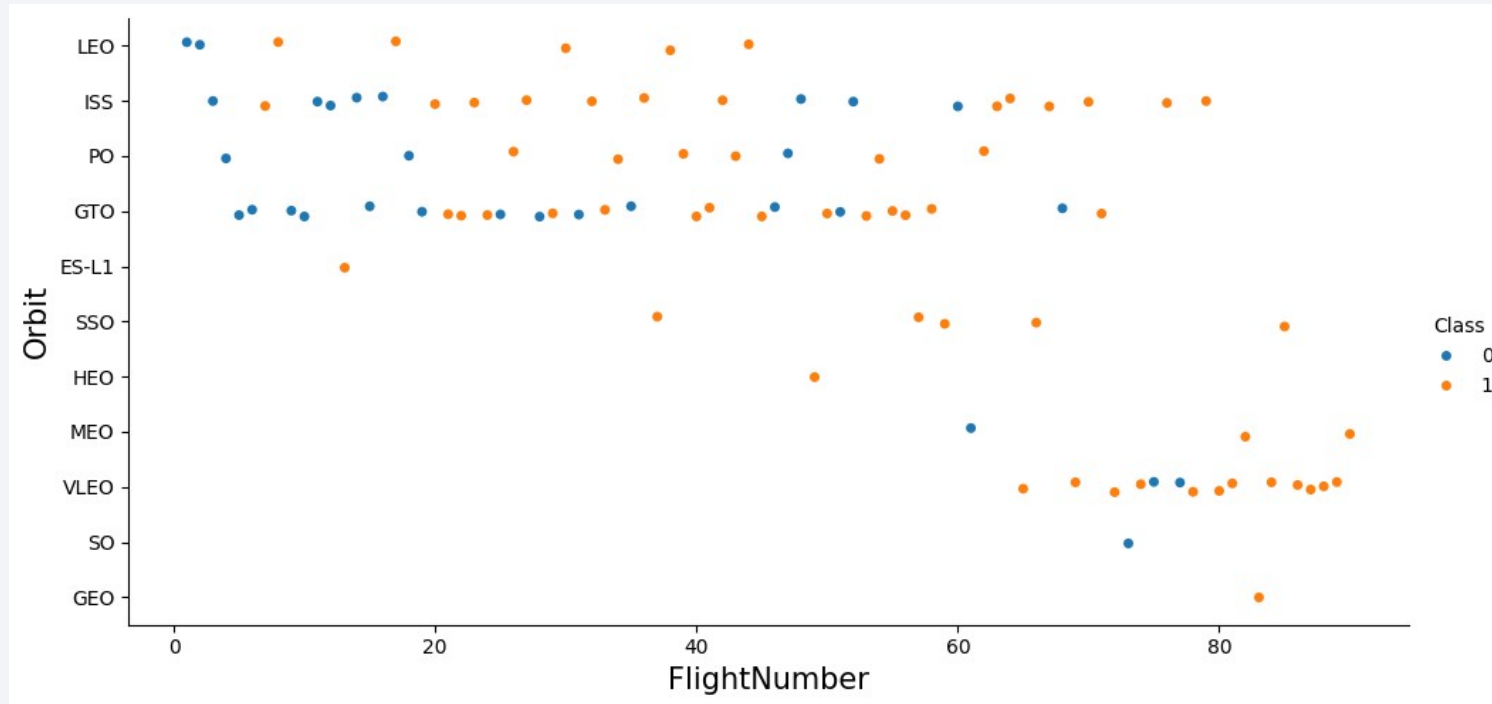
Bar chart for the success rate of each orbit type



ES-L1, GEO, HEO and SSO have 100% success rate. GTO has one of second lowest success rate indicated that this orbit is challenging for a successful mission. Orbit SO has the no success suggesting that no mission used that orbit or that this orbit is complex or close to impossible.

Flight Number vs. Orbit Type

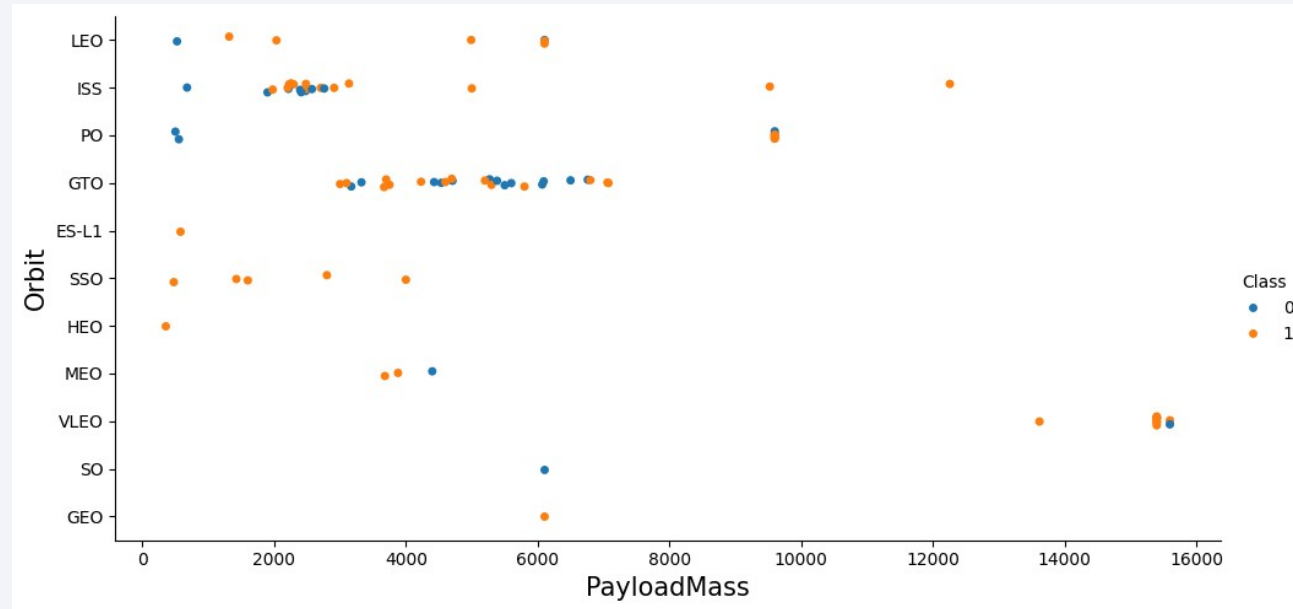
Flight number vs. Orbit type



Yellow represents success while blue represents failure. The graph shows that early flights with orbit type ISS, PO and GTO had mixed results. There's a preference with VLEO for later flights. Orbits with the greatest success has the least amount of missions.

Payload vs. Orbit Type

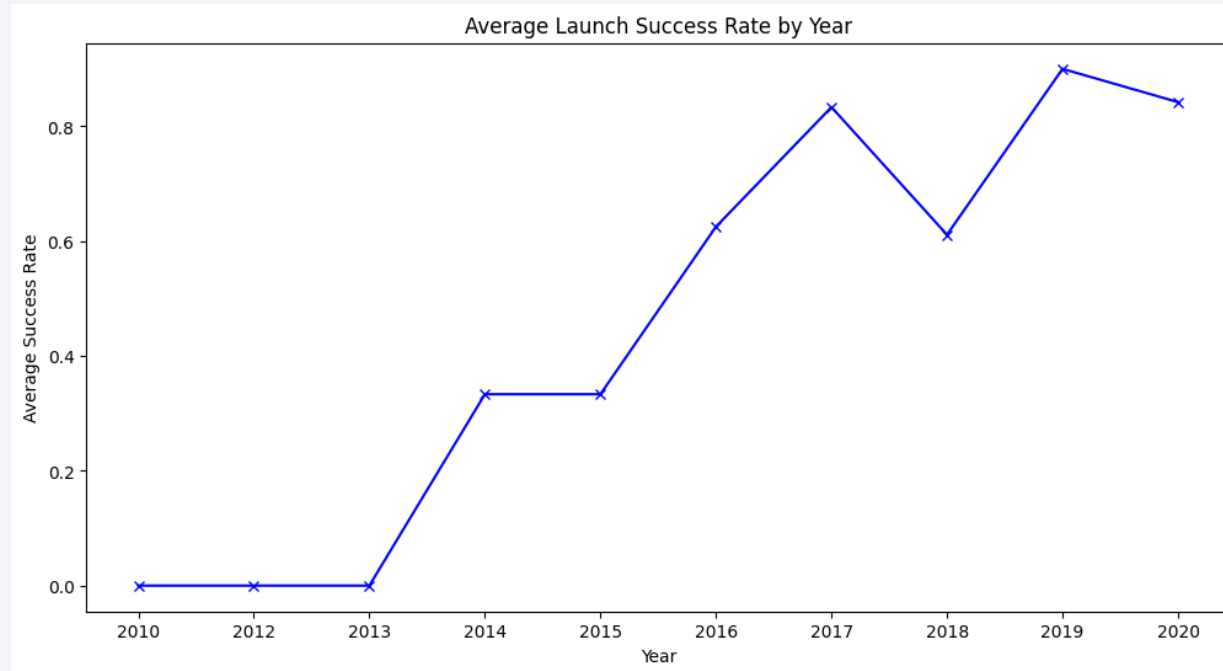
Payload vs. Orbit Type



Yellow represents success while blue represents failure. The graph shows successful orbit VLEO only has payload mass values in the higher end of the range. SSO, LEO and MEO seems to have only low payload mass 6000 kg and less.

Launch Success Yearly Trend

Yearly Average Success Rate



Yellow represents success while blue represents failure. The graph shows that success rate increased over time since 2013 with a slight dip in 2018.

All Launch Site Names

Find the names of the unique launch sites

```
%sql SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE;  
✓ 0.0s  
* sqlite:///my\_data1.db  
Done.  
  


| Launch_Site  |
|--------------|
| CCAFS LC-40  |
| VAFB SLC-4E  |
| KSC LC-39A   |
| CCAFS SLC-40 |


```

The above show the unique launch sites from the database. CCAFS LC-40 and CCAFS SLC-40 are likely the same launch sites misspelled.

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with `CCA`

```
%%sql SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

✓ 0.0s Python

* [sqlite:///my_data1.db](#)

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Calculate the total payload carried by boosters from NASA

```
%%sql SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE  
WHERE Customer = 'NASA (CRS)';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

SUM(PAYLOAD_MASS_KG_)

45596

The above show that a total 45596 kg of payload mass was sent to NASA (CRS). CRS is for Commercial Resupply Services which suggest that these payloads were resupply missions to the International Space Station.

Average Payload Mass by F9 v1.1

Calculate the average payload mass carried by booster version F9 v1.1

```
%%sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE  
WHERE Booster_Version LIKE 'F9 v1.0%';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

AVG(PAYLOAD_MASS_KG_)

340.4

First Successful Ground Landing Date

Find the dates of the first successful landing outcome on ground pad

```
%%sql SELECT MIN(DATE) FROM SPACEXTABLE  
WHERE Landing_Outcome LIKE '%Success%';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)
Done.

MIN(DATE)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%%sql SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;
✓ 0.0s

* sqlite:///my\_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The above show a query returning four booster versions that had successful drone ship landings with a payload mass between ranging from 4000 and 6000 non-inclusively.

Total Number of Successful and Failure Mission Outcomes

Calculate the total number of successful and failure mission outcomes

```
%%sql SELECT Mission_Outcome, COUNT(*) AS TOTAL FROM SPACEXTABLE
GROUP BY Mission_Outcome;
✓ 0.0s

* sqlite:///my\_data1.db
Done.
```

Mission_Outcome	TOTAL
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The above shows the count for each mission outcome. Success appears multiple time due to trailing spaces and different inputs used for success.

Boosters Carried Maximum Payload

List the names of the booster which have carried the maximum payload mass

```
%%sql SELECT Booster_Version, Payload_Mass_KG_ FROM SPACEXTABLE  
WHERE Payload_Mass_KG_ = (SELECT MAX(Payload_Mass_KG_) FROM SPACEXTABLE);
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%%sql SELECT CASE substr(Date, 6, 2)
WHEN '01' THEN 'January'
WHEN '02' THEN 'February'
WHEN '03' THEN 'March'
WHEN '04' THEN 'April'
WHEN '05' THEN 'May'
WHEN '06' THEN 'June'
WHEN '07' THEN 'July'
WHEN '08' THEN 'August'
WHEN '09' THEN 'September'
WHEN '10' THEN 'October'
WHEN '11' THEN 'November'
WHEN '12' THEN 'December'
END AS Month_Name, Landing_Outcome AS Failure_Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE DATE LIKE '2015%' AND Landing_Outcome LIKE 'Failure (drone ship)';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Month_Name	Failure_Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The above shows the failed landing outcomes for drone ship along with the month, booster version and launch site.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%%sql SELECT Landing_Outcome, COUNT(*) AS outcome_count FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY outcome_count DESC;
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

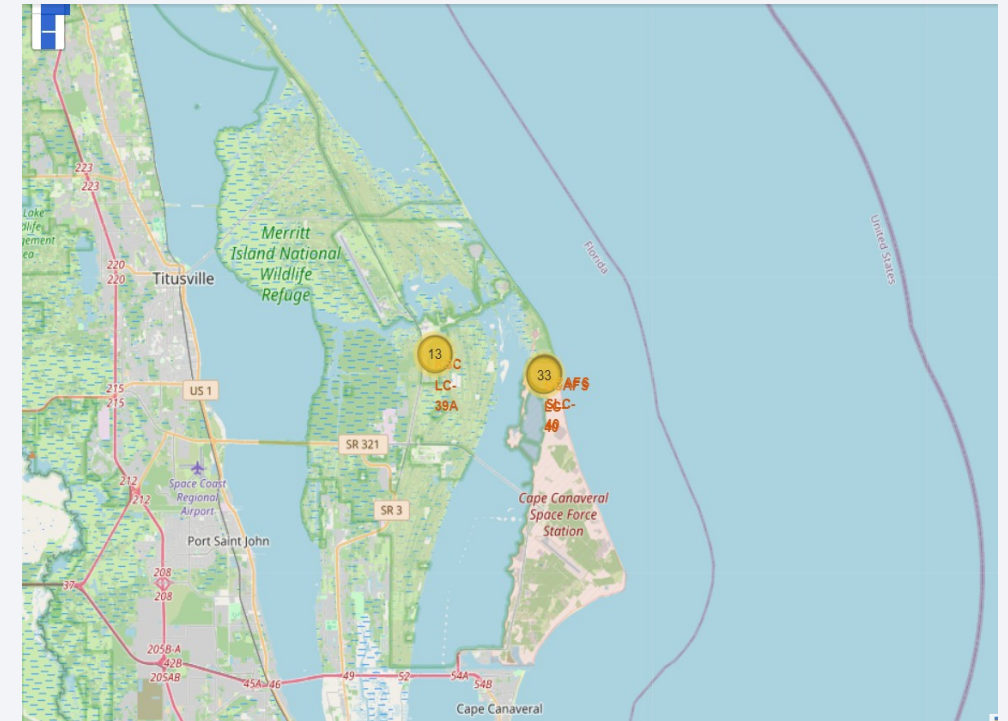
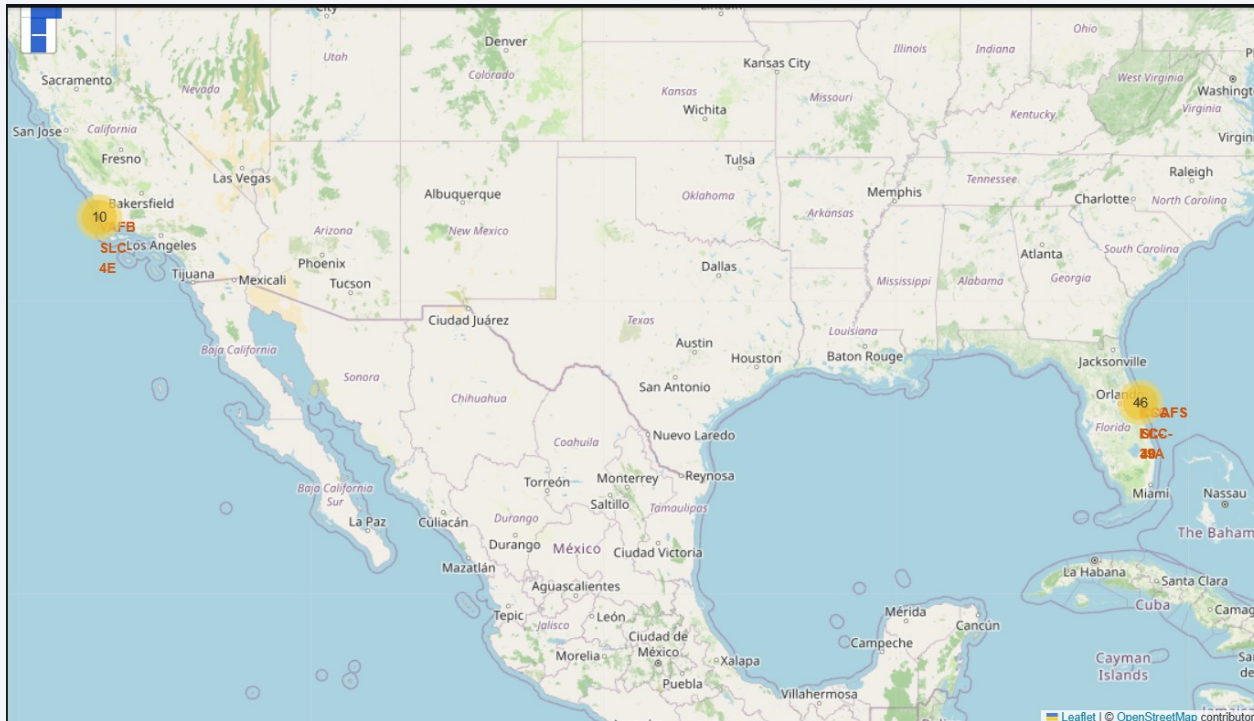
The above shows the landing outcome counts for all landing outcomes between 2010-06-04 and 2017-03-20.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear view of the Earth's horizon and the surrounding space.

Section 3

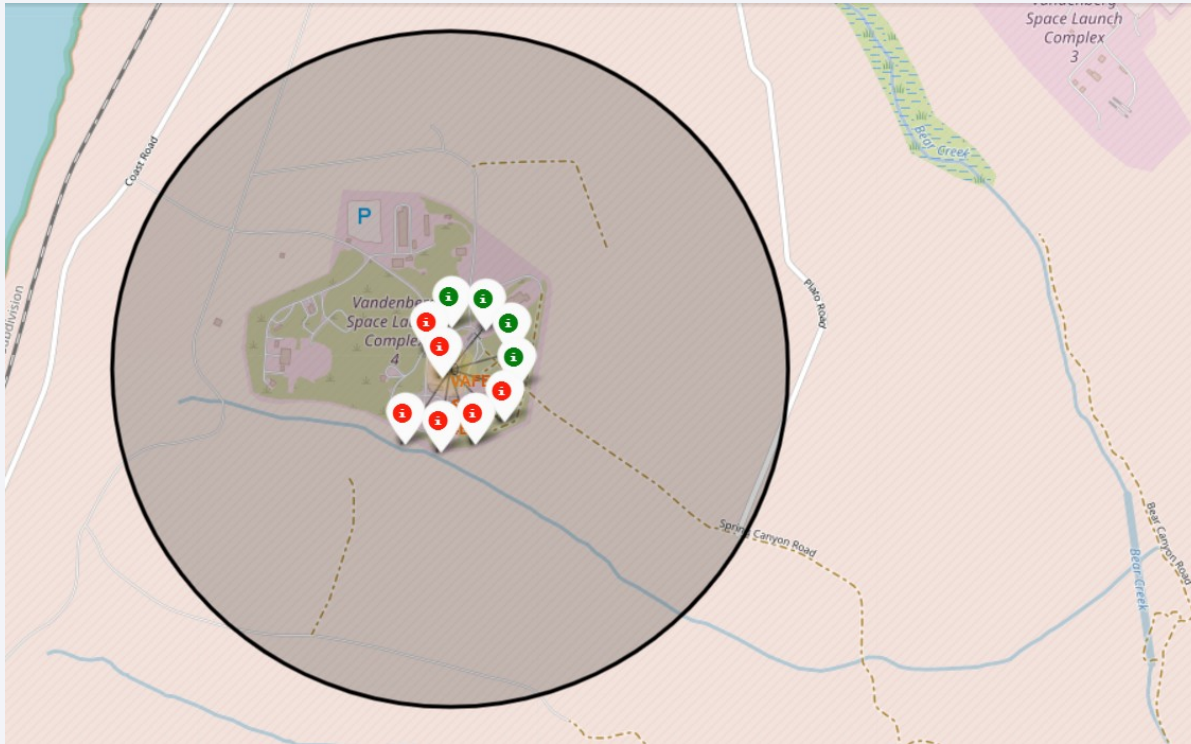
Launch Sites Proximities Analysis

Launch Site locations



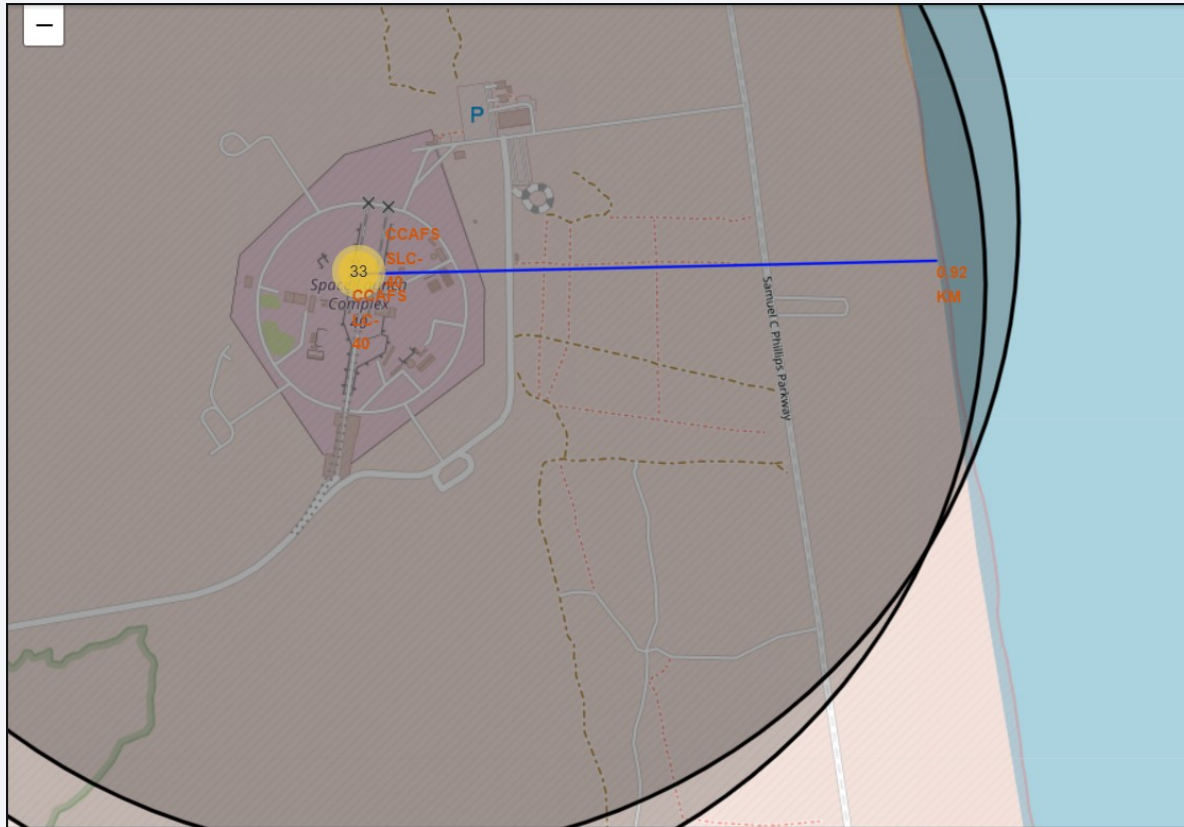
The above shows all the launch site. The left image shows that Florida has 2 launch sites.

Launch Site Marker

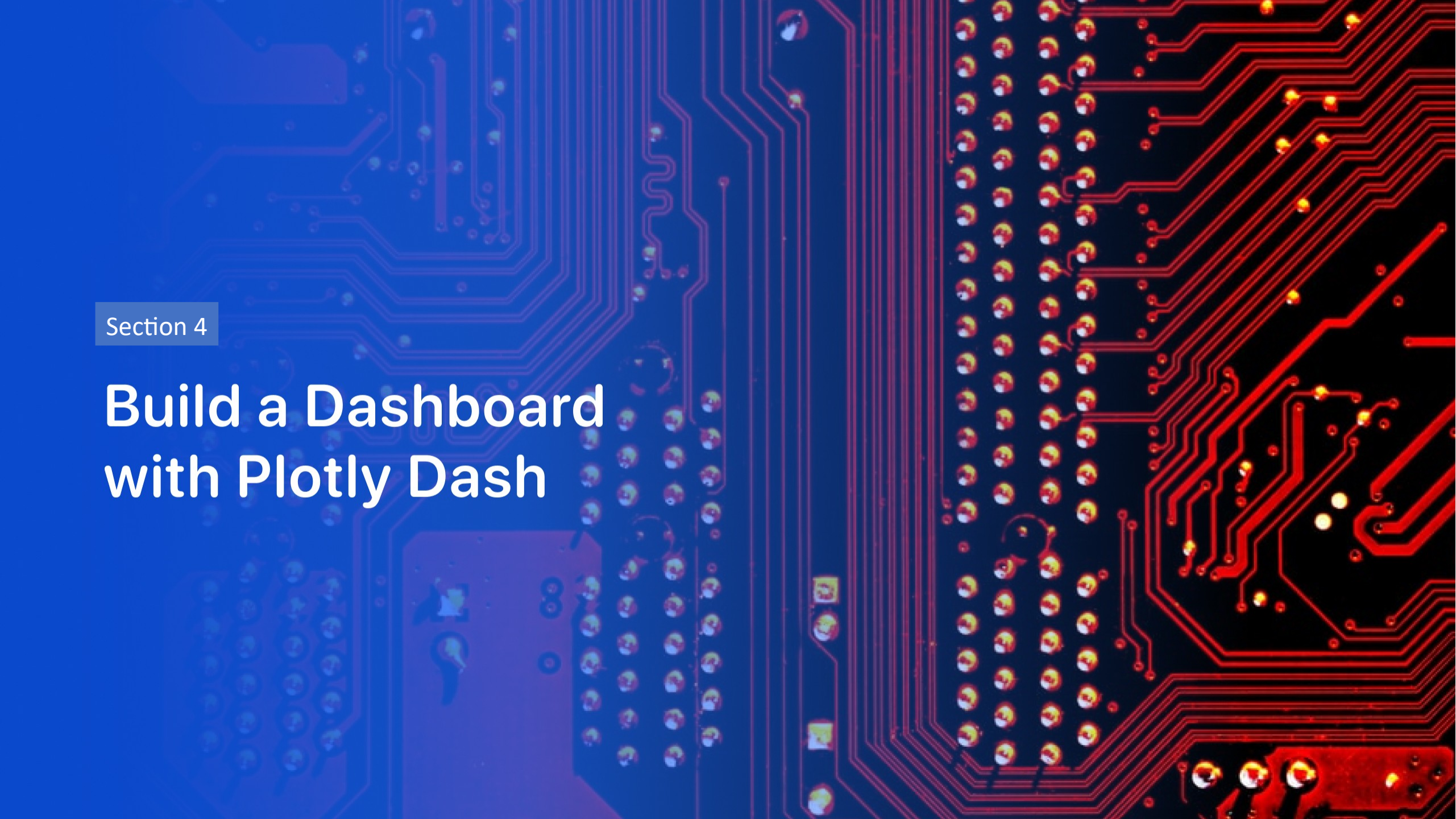


The above indicates the successful and failed missions from the launch site.

Launch Site Proximity



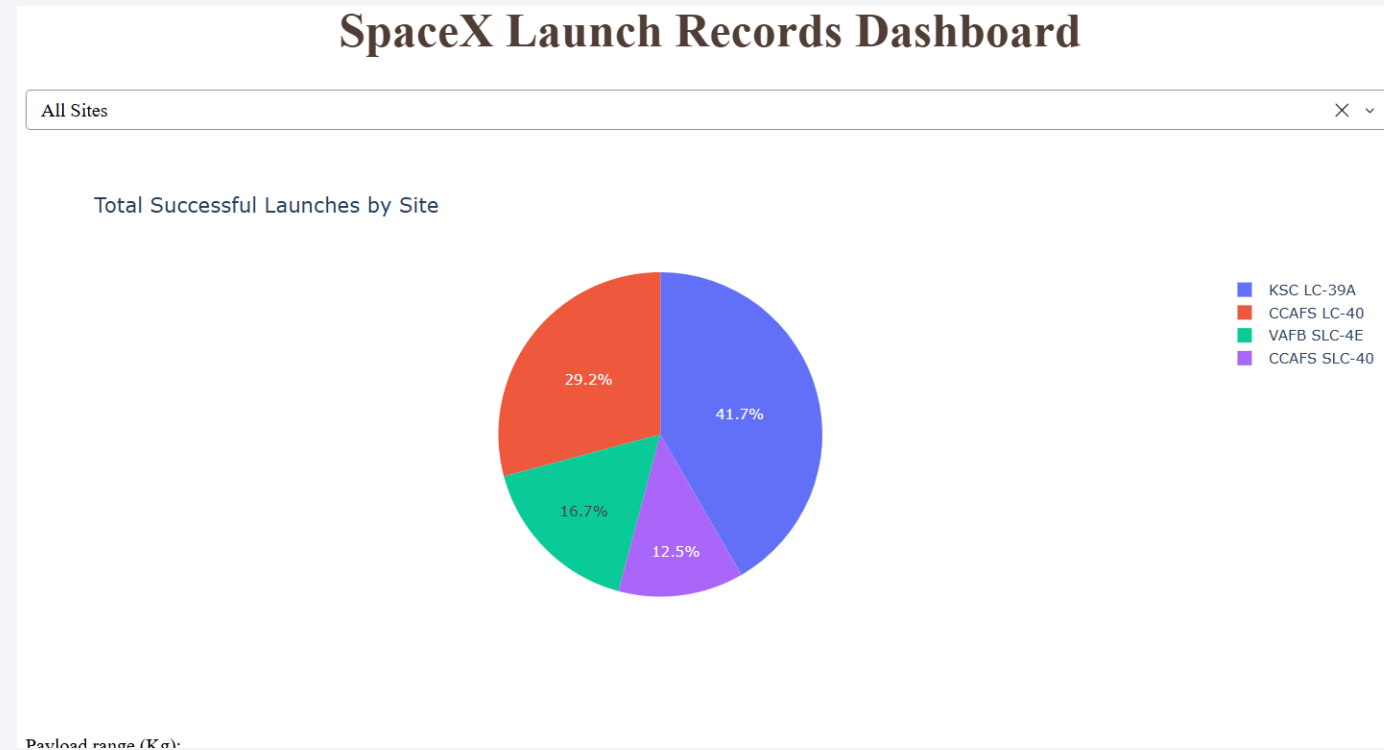
The above indicates a launch site with it's calculated distance from the coastline.



Section 4

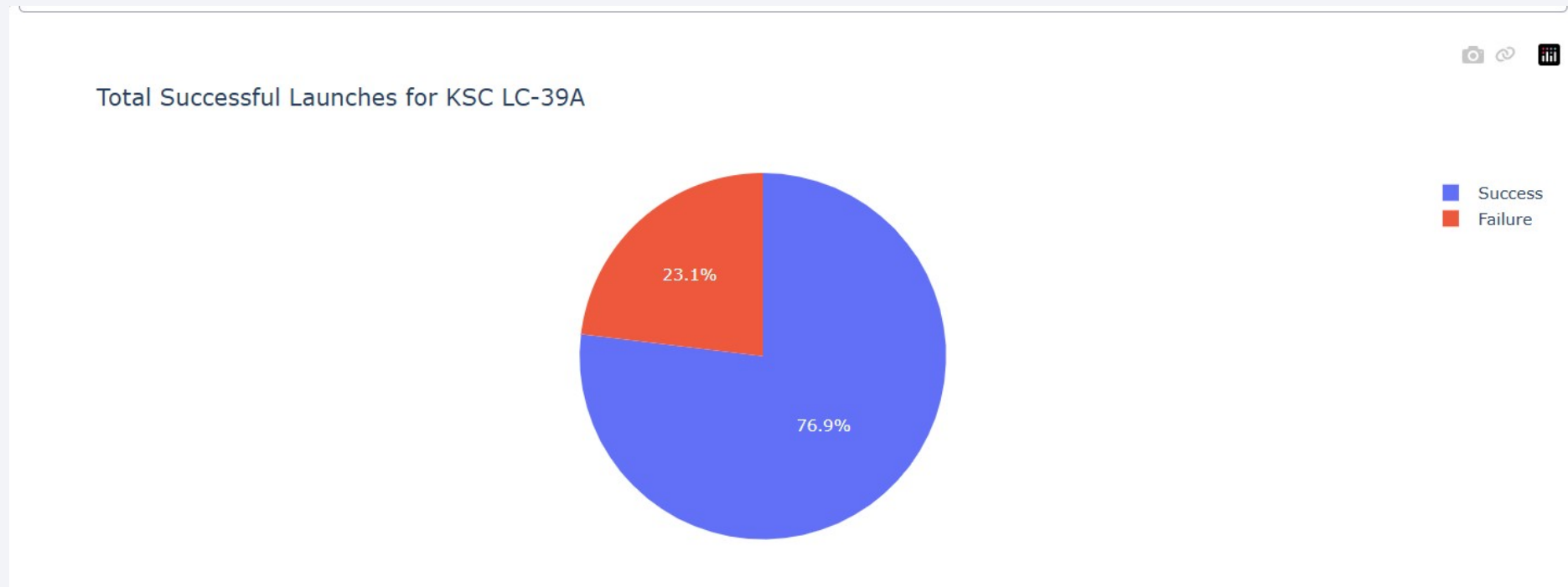
Build a Dashboard with Plotly Dash

Successful Launch by Sites



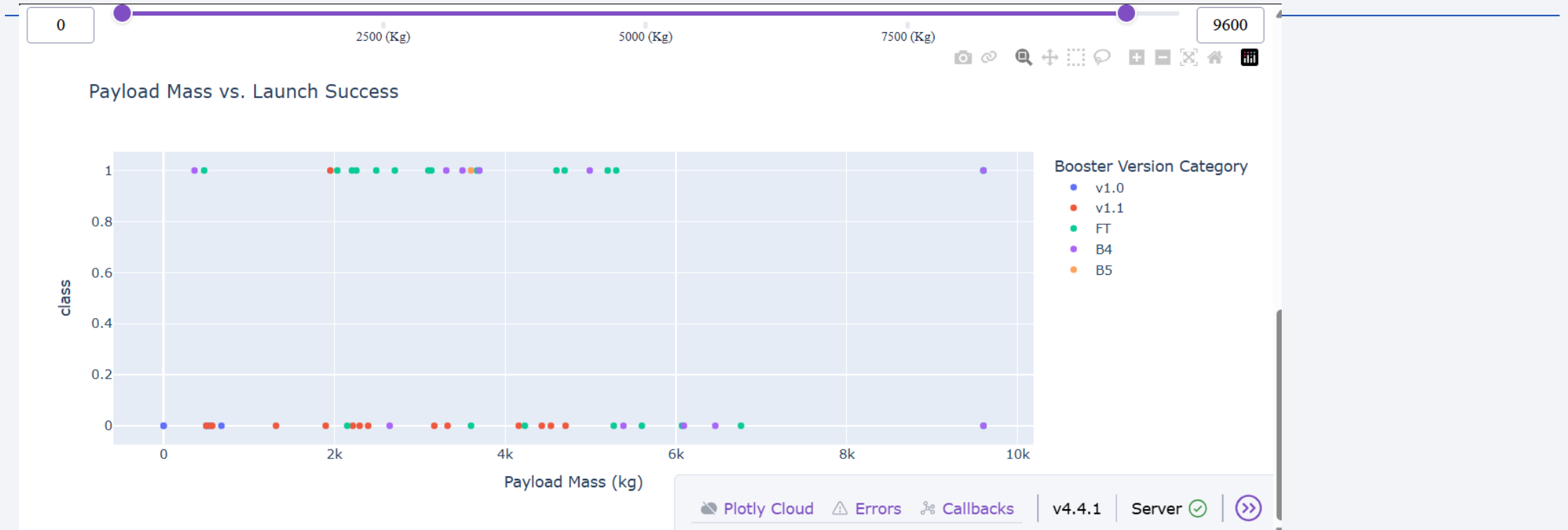
The distribution shows that KSC LC-39A launch site has the greatest successful launches. This suggest that KSC LA-39A is the most reliable site for SpaceX launches.

Success Rate Distruction for KSC LC-39A



The above chart shows the portion of the KSC LC-39A launches that's successful vs the failures.

Payload vs. Launch Outcome for all launch Sites



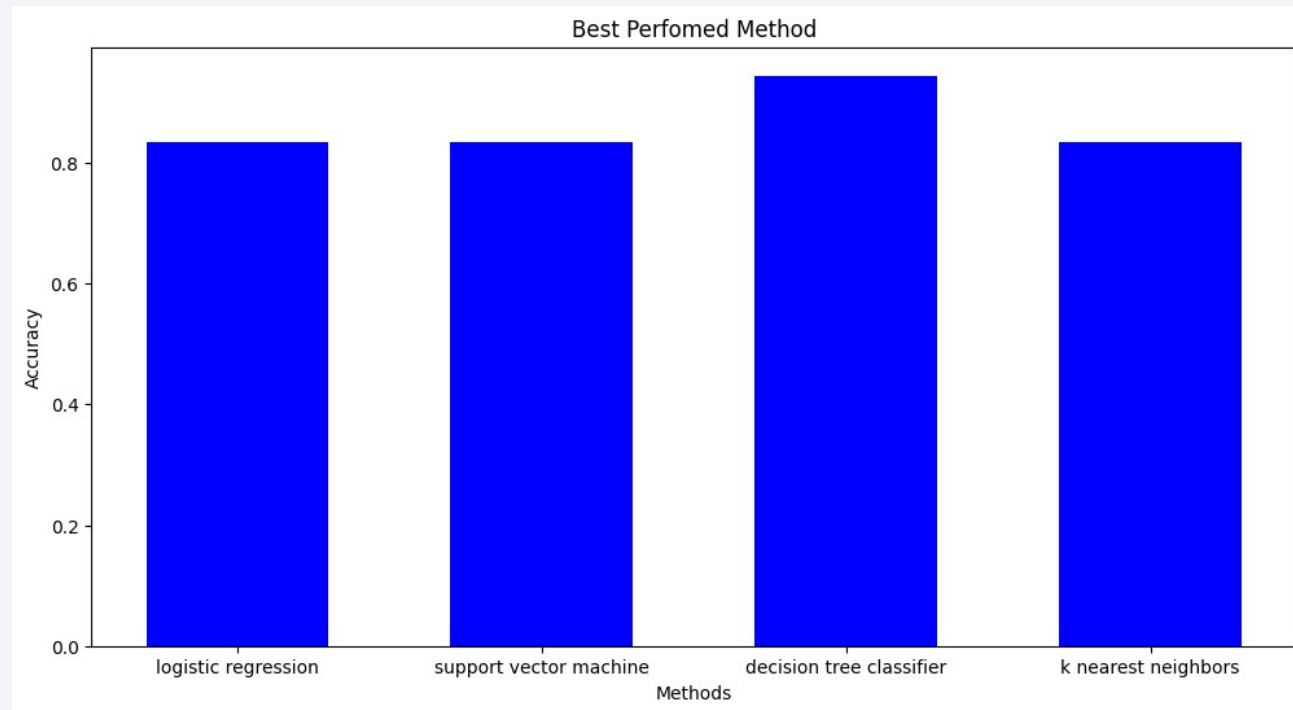
For the chart displayed above class 1 represents a successful outcome while 0 represents a failure. The FT Booster Version Category seems to have the more successful outcomes.



Section 5

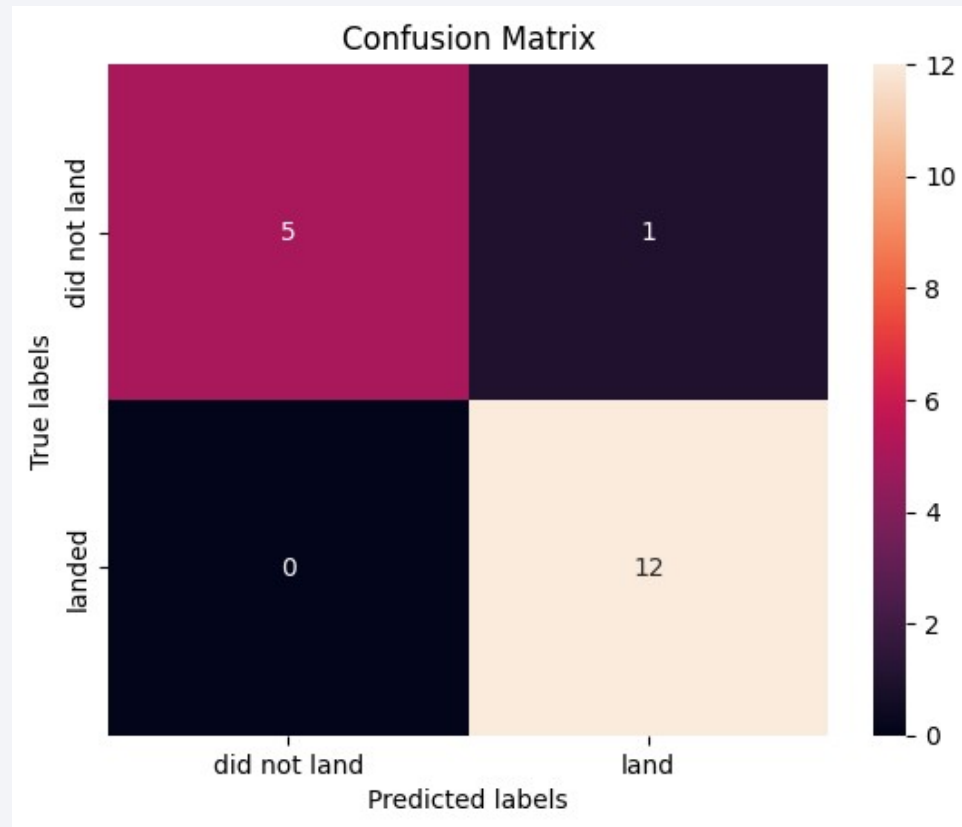
Predictive Analysis (Classification)

Classification Accuracy



The above chart shows that the Decision Tree Classifier has the highest accuracy which shows that it is the best performing method.

Confusion Matrix



The above shows the confusion matrix for the Decision Tree Classifier. The predicted and true values for mission outcomes that have landed is high correlation with each other, this high correlation is also seen in the result for did not land.

Conclusions

- Falcon 9 landing success improved substantially over time, surpassing 80% by 2020.
- KSC LC-39A recorded the highest share of successful launches at 41.7%.
- Launch site and booster version were stronger predictive factors than payload mass alone.
- The Decision Tree classifier achieved the highest accuracy at 94.44%.
- Combining data engineering, EDA, visualization, and machine learning provided a comprehensive view of Falcon 9 landing performance.
- The analysis demonstrates that historical launch data can support more informed predictions of future landing outcomes.

Appendix

GitHub: <https://github.com/RogerIO-git/IBM-Coursera-Capstone>

Thank you!

